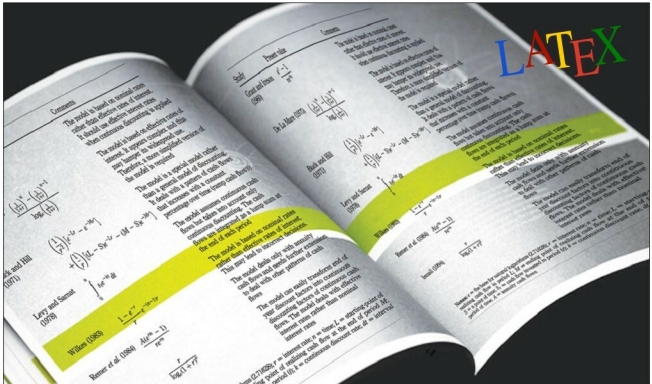


Explore LaTeX and Syntax Highlighting with Scilab

This article, the last of a four-part series on Scilab, takes a look at LaTeX and syntax highlighting with Scilab. It is aimed at readers interested in technical computation.



Scilab (<http://www.scilab.org>) is open source software for numerical computation with a syntax similar to MathWorks' MATLAB. The first part of this article is about the use of LaTeX with Scilab, and the second concerns the keywords and how to use them to build a syntax highlighting file for GNU Emacs and Vim. All the code presented here has been tested with Scilab 5.5.1 under Linux Mint 17 Xfce. Since the full code is too long for a printed article, it's freely (as in freedom and in beer) available at <https://github.com/astonje/scilab>

LaTeX on a plot

Version 5.2 is the first version of Scilab in which it is possible to use LaTeX. The rendering engine is based on JLaTeXMath (see Reference 1), a fork of JMathTeX (Reference 2) which is a Java library for the mathematical mode of LaTeX. It's not necessary to install LaTeX to use it with Scilab. Let's consider the plot shown in Figure 1, which shows two bell-shaped curves: the witch of Agnesi

(Reference 3) and the function of Gauss. This plot has seven textual parts:

1. **Title.** The syntax is: `title("$this-is-the-title$", "color", "color-name")`. First, it's necessary to consider that a text written in LaTeX must be included between two '\$' symbols. In this example, the full title is written in the colour blue. Two or more colours are possible by adding the LaTeX command `\textcolor`. On the plot, I can see the first part written in blue and the second in black. If I put the mouse on the LaTeX code, SciNotes, the built-in editor shows, an all-black preview (Figure 2), while the blue part is not visible.
2. **X axis label.** The syntax is very easy: `xlabel("$this-is-the-label$")`.
3. **Y axis label.** The syntax with `ylabel` is the same as in the X axis.
4. **X ticks labels.** In this example, labels are defined as strings in the vector `labx`. Using `x_ticks.labels` within a loop, I can put each label on the plot. The syntax is: